

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Sixth Semester B.E. (E&E) (Credit System) Degree Examinations

May - June 2023

20EEE102 – SWITCHED MODE POWER CONVERTER

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I

Marks BT* CO* PO*

1. a) Explain the working of a Buck converter under continuous conduction mode. Draw the input output voltage, inductor current and capacitor ripple waveforms. Give the design details of the filter components used in the circuit. 10 L*2 1 1,2
- b) Design a forward converter such that the output is 5V when the input is 170 V. The output current is 5 A. The output voltage ripple must not exceed 1%. Choose the transformer turns ratio, duty ratio, and switching frequency. Choose L_s such that the current in it is continuous. Include the ESR when choosing a capacitor. Use $r_C = 10^{-5}/C$. Consider $f = 300\text{kHz}$. Draw the circuit diagram of the designed converter 10 L4 2 2,3
2. a) Design a boost converter that will have an output of 30 V from a 12V source. Design for continuous inductor current and an output ripple voltage of less than one percent. The load is a resistance of 50Ω . Assume ideal components for this design with switching frequency of 25kHz. Draw the circuit diagram. 10 L4 1 2,3
- b) Explain working of full bridge and half bridge converter with the help of neat circuit diagram and relevant waveforms. Give the equations for the output voltage, switch ratings in each case. 10 L2 2 1,2
3. a) Explain the working of a push pull converter with neat circuit diagram and relevant waveforms. Obtain the equation for output voltage. 10 L2 1 1,2
- b) Explain the working of Flyback Converter with neat circuit diagram and relevant waveforms. With suitable assumptions derive the equation for output voltage. Explain the design procedure to be followed. 10 L2 2 1,2

Unit – II

4. a) Explain the working of Single-Phase Full-Bridge Inverter controlled by Sinusoidal PWM with bipolar switching scheme with necessary circuit diagram and waveforms. Comment on the harmonics in the output voltage. 10 L2 3 1,2
- b) Explain the concept of Zero Voltage Switching. How it can be realized for a Buck converter. Explain with relevant circuit diagram and waveforms. 10 L2 4 1,2
5. a) Explain the PWM technique used in three-phase voltage source inverters. Explain the linear modulation and also indicate the harmonic spectrum for unipolar PWM with $m_f = 15$. 10 L2 3 1,2

P.T.O.

- b) For a ZCS $V_s=12\text{ V}$, $C_r=0.1\mu\text{F}$, $L_r=10\mu\text{H}$, $I_o=1\text{A}$, $f_s=100\text{ kHz}$ (i) Determine the output voltage of the converter. (ii) Determine the peak current in L_r and the peak voltage across C_r . (iii) What is the required switching frequency to produce an output voltage of 6 V for the same load current? Draw the circuit diagram. 10 L3 4 2,3
6. a) Explain the working of Single-Phase Full-Bridge Inverter controlled by Sinusoidal PWM with unipolar switching scheme with necessary circuit diagram and waveforms. Also Explain the square wave operation of single-phase inverter circuit. 10 L2 3 1,2
- b) With a neat circuit diagram explain the series resonant DC-DC converter. Draw the waveforms considering $\omega_s > \omega_0$. Derive the output voltage equation. 10 L2 4 1,2
- Unit – III**
7. a) Explain the types and sources of disturbances. Explain different power conditioners used to suppress these disturbances. 8 L2 5 1,2
- b) Explain in detail the design of high frequency transformer for isolated switched mode power converters. How magnetizing inductance can be added in the primary winding while designing such transformers? 12 L2 5 1,2
8. a) Design an inductor for the buck converter with following parameters. $V_o=5\text{V}$, $I_o=5\text{A}$, switching frequency= 40kHz , input voltage $V_s=12\text{V} \pm 10\%$ considering ripple in inductor current equal to 10% of output current. Calculate area product, number of turns, air gap length, wire cross sectional area. 12 L3 5 2,3
- b) With a block schematic, explain the different components an Uninterruptable Power Supply (UPS). 8 L2 5 1,2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
